

Controller and Physical Interface Solutions for Off-chip Memories

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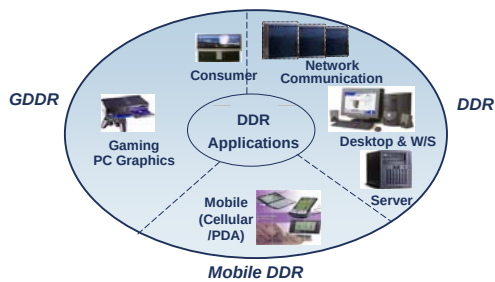


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Wide Variety of DDR Applications



Mainstream devices moving towards high performance
Cell phone #1 new market for DDR

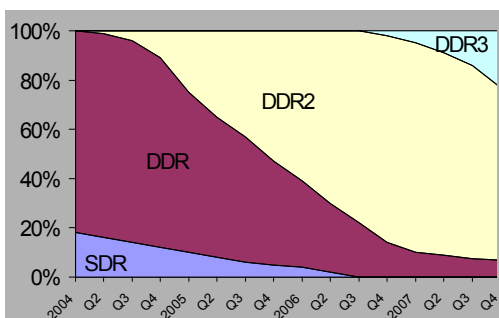


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Long Term DRAM Market Trend



Source: Denali and Hynix (2004)



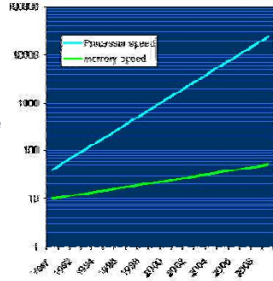
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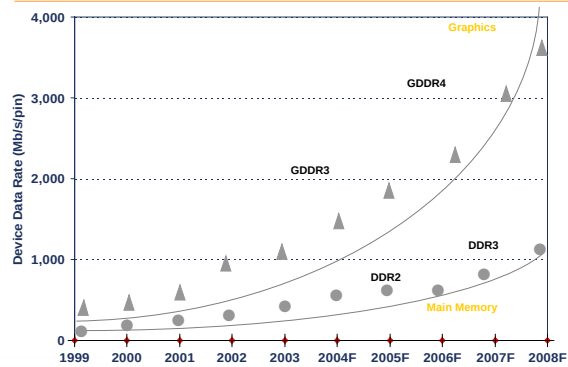


Performance: Processor-Memory Gap

- Speed of external memory does NOT keep pace with SoC processors
 - Increasing memory array size
 - Timing limitations between separate ICs
- Bandwidth gap
 - Reduced through double data-rate clocking (DDR)
 - Boosted through wider memory buses
 - At the cost of I/O
- Latency gap remains
 - Primarily initial access latency

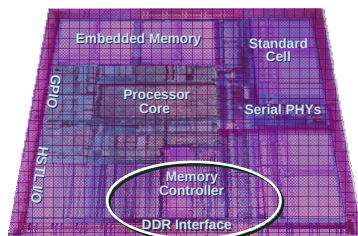


Increasing Device Bandwidth



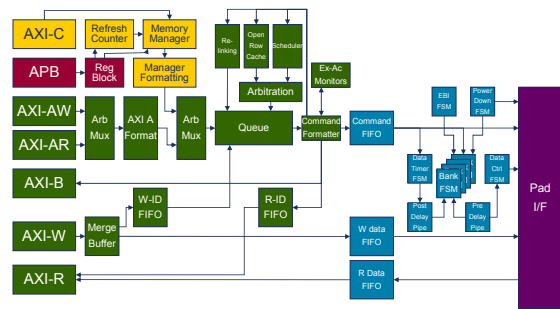
ARM® SoC IP Library

Pervasive SoC IP

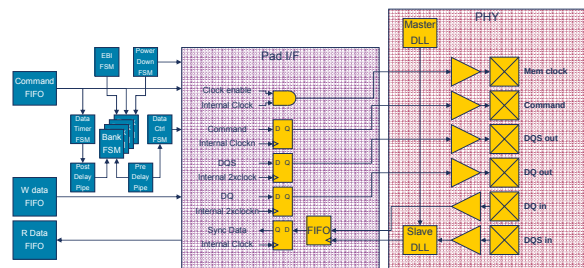


- Library platforms
 - Standard cells
 - Embedded memory
 - I/O functionality
- High-speed interfaces
 - DDR
 - Serial PHYs

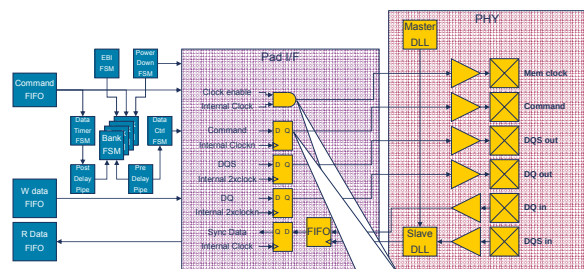
PL340 – Functional Block Diagram



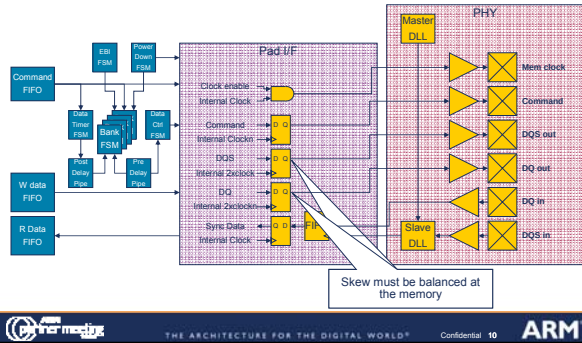
PL340 – Functional Block Diagram



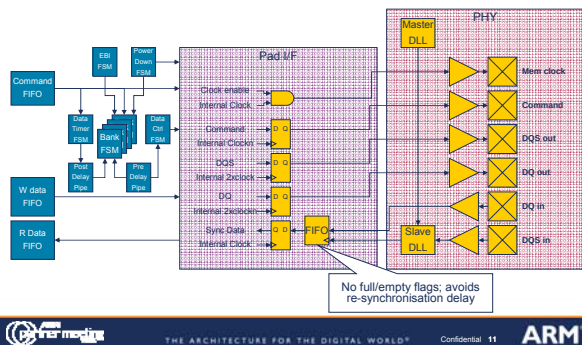
Command Skew



Data Output Skew

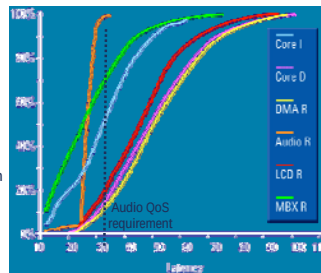


Read Data Skew



PrimeCell® Dynamic Memory Controllers

- Extensive benchmarking throughout development
 - SystemC TLM model
 - RTL simulation and FPGA prototype
- 2x performance for half the gate count of previous generation
 - Example : 32-bit mobile DDR
 - 166 MHz clock rate (333 MHz data)
 - Up to 1GB/sec bandwidth
- Efficient architecture gives
 - Low gate count
 - Typically 90K gates
 - Low latency for critical masters
 - Via configurable quality of service (QoS) requirement



PrimeCell Dynamic Memory Controllers

- Devices supported by PL340 controller
 - SDRAM – including low-power SDRAM
 - DDR SDRAM
 - Mobile DDR SDRAM
 - Up to 166MHz on 0.13μ process (SageX-HS)
- Devices supported by PL341 DDR2 controller
 - High-performance interface to DDR2
 - Supporting memory and AXI frequencies up to 400MHz (800MHz data) on a 90nm process (SageX-HS)



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PrimeCell Dynamic Memory Controllers

- Configurable for:
 - Memory width (16-bit, 32-bit and 64-bit)
 - AMBA® 3 AXI interface width (32-bit, 64-bit and 128-bit)
 - 1 to 4 memory chips with 2 or 4 banks per chip
- Data buffer FIFO depths
 - Outstanding transaction queue depth
- Effective power management through:
 - AXI-C power control interface
 - Software controlled stand-by
 - Hardware controlled stand-by
 - RealView® SoC Designer model
 - High-performance model enables:
 - Performance evaluation at the system level
 - Optimal configuration selection

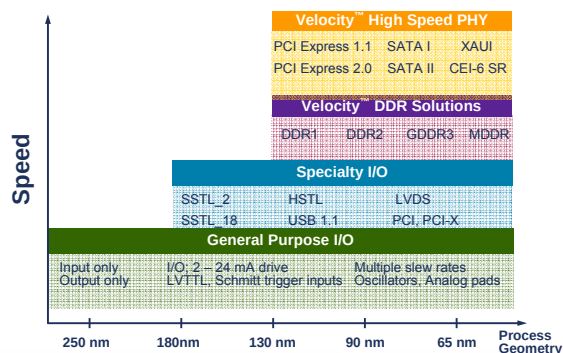


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I/O & High-Speed Interface Product Portfolio

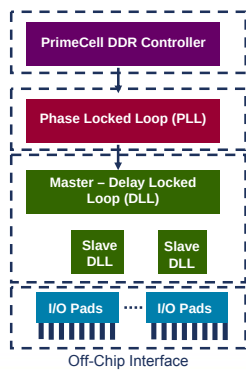


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Flexible DDR Solution



- ARM PrimeCell DDR controllers:
 - AXI DDR/MDDR (PL340) - Available Now
 - AXI DDR2 (PL341) - Available Oct '06
 - AHB DDR (PL244 & PL245) - Available Now
- Flexible implementation to optimize solution area, power, performance
- Switchable architecture allows DDRI, DDRII, GDDR3
- Up to 1.8Gb/s(GDDR3) data throughput
- Backwards compatible with DDR (2.5V I/O drive) using lower drive strengths
- Contains dynamic calibrator for on-die termination and pull-up/pull-down driver network for PVT variations



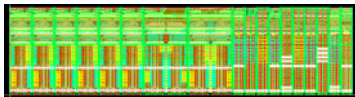
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DDR Silicon PHY IP Components

- IO cell set
 - Combo(DDR1/2) or dedicated set(GDDR3)
 - Power / ground
 - Decap cells
- DLL master and slave
- Deskew PLL
- ARM Fabric DDR Controller's also available:
 - ARM PL340 – Mobile DDR
 - ARM PL244 and PL245 – DDR1



DDR Combo pad set



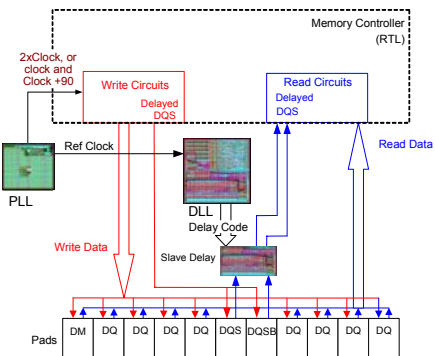
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DDR Interface Components

8-Bit Module



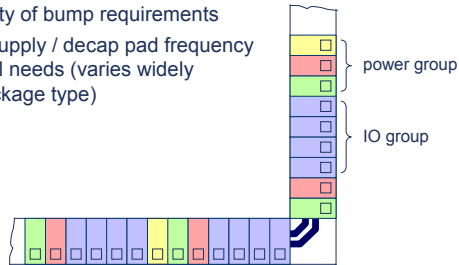
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IO Placement Flexibility

- No hardened PHY
- Problem of the chip corner
- Variability of bump requirements
- Match supply / decap pad frequency to actual needs (varies widely with package type)

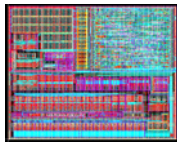
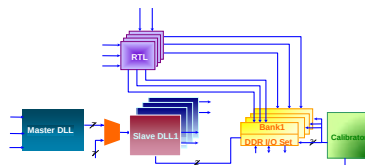


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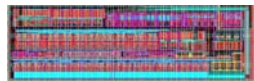
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DLL Use in DDR Interface PHY



Master DLL



Slave DLL

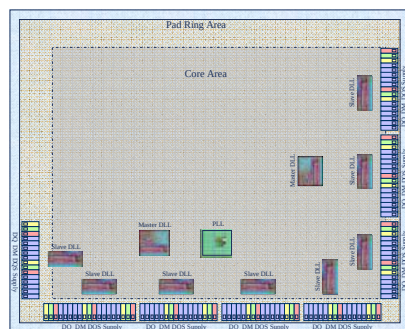


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Layout Example of a 64-bit DDR Bus



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DDR Solutions for 90nm

Standard/App		Soft IP	Hard IP	
		Controller	Analog	I/Os
DDR I 2.5V <i>Up To 400Mb/s</i>	SDRAM	ARM PL244 ARM PL245 ARM PL340	800Mbs DLL 533MHz PLL	DDR1/2
DDR II 1.8V <i>Up To 800Mb/s</i>	SDRAM/ RLDRAM	ARM PL341 Q406 3 rd Party	800Mbs DLL 533MHz PLL	DDR1/2
GDDR III 1.8V <i>Up To 1.6Gb/s</i>	SDRAM/ RLDRAM	3 rd Party	1600Mbs DLL 800MHz PLL	GDDR3



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ARM

DDR Solutions for 130nm

Standard/App		Soft IP	Hard IP	
		Controller	Analog	I/Os
DDR I 2.5V <i>Up To 400Mb/s</i>	SDRAM	ARM PL244 ARM PL245 ARM PL340	800Mbs DLL 533MHz PLL	DDR1/2 Mobile DDR
DDR II 1.8V <i>Up To 800Mb/s</i>	SDRAM/ RLDRAM	ARM PL341 Q406 3 rd Party	800Mbs DLL 533MHz PLL	DDR1/2
GDDR III 1.8V <i>Up To 1.4Gb/s</i>	SDRAM/ RLDRAM	3 rd Party	1400Mbs DLL 800MHz PLL	DDR1/2/GDDR3

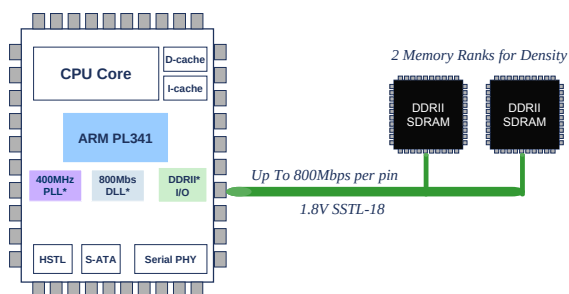


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DDR II System Example



*Standard IP 533MHz PLL, 800Mbs DLL, DDR1/2 I/O

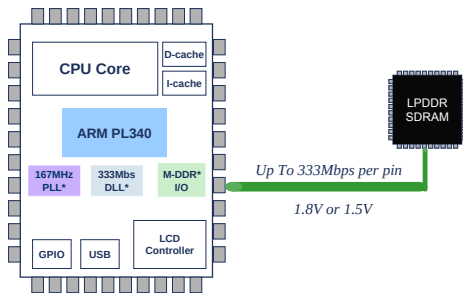


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Mobile DDR System Example



*Standard IP 533MHz PLL, 800Mbps DLL, M-DDR I/O

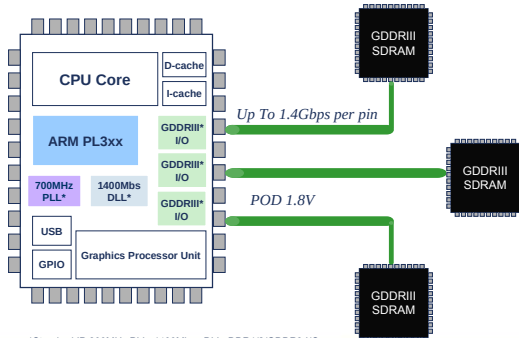


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GDDRIII System Example



*Standard IP 800MHz PLL, 1400Mbps DLL, DDR1/2/GDDR3 I/O

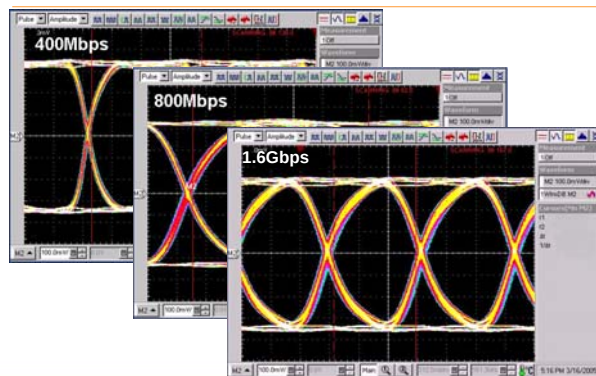


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Proven In Silicon



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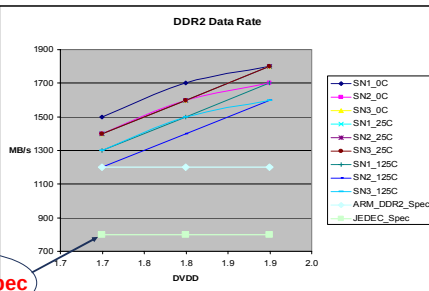
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Silicon Proven DDR1/2 IP 90nmG

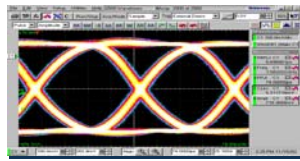
**TSMC
Quality
Criteria**

JEDEC Spec



In-house Silicon Validation

- Design reliability
 - Silicon validation
 - Noise analysis
 - Performance above specification
 - Manage SSO effects



- Validation
 - Key to minimize risk and time to market
 - Silicon characterization
 - Validating high-speed memory interfaces can be costly

ARM Delivers Superior Value

- Superior technical solution
 - Flexible, backward-compatible
 - Performance, power, area – smallest, tightest pad pitch
 - Modular: consistent form factor across mixed-signal products
 - Low jitter for compliance with tight timing of DDR specification
- Cost savings
 - Smallest size
 - Foundry portability
- Capabilities and experience
 - R&D team, support, silicon, lab
- Documentation
 - White paper reference implementation (Operatic PL340/Mobile DDR)
- Manufacturability
 - History of high-volume production

